STUDENT PERFORMANCE ANALYSIS USING EXCEL AND PYTHON

Name: Tupakula Suchithra

Course: BTech (ECE)

Date: April 2026

1. Introduction

This project focuses on analyzing student academic performance using Excel and Python. The aim is to understand how data can be used to evaluate student performance, calculate key metrics, and visualize results effectively.

2. Objective

- To analyze student marks data
- To calculate total and average scores
- To identify top and weak performers
- To visualize performance using charts

3. Tools Used

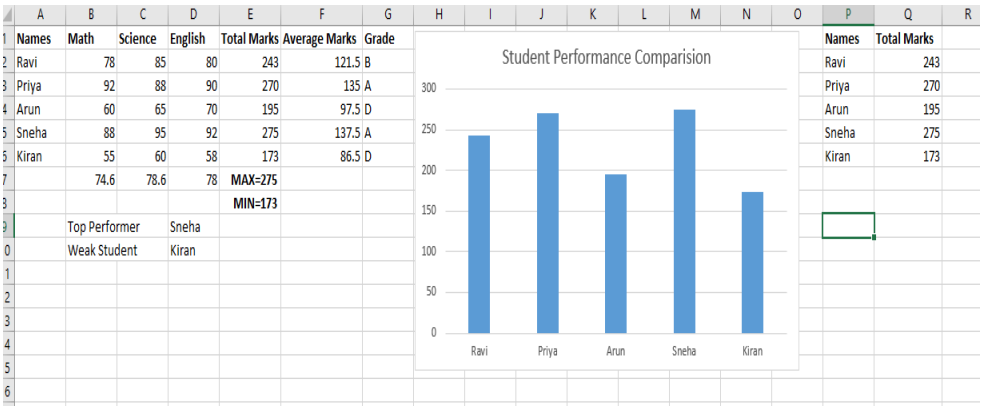
- Microsoft Excel
- Python
- Pandas
- Matplotlib

4. Dataset Description

The dataset contains student names and their marks in three subjects: Math, Science, and English. The data was manually created for analysis purposes.

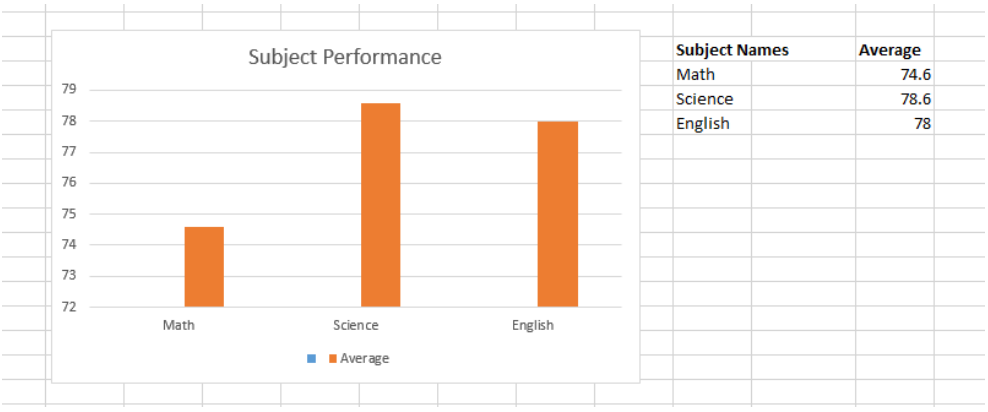
5. Methodology

- Data was organized in tabular format
- Total and average marks were calculated
- Top and weak performers were identified
- Charts were created to visualize performance



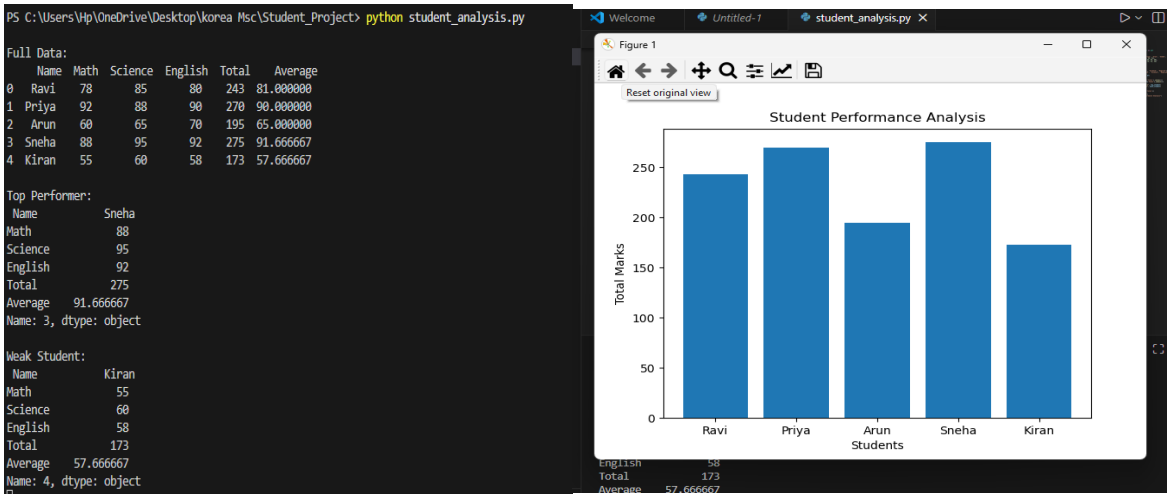
Names	Math	Science	English	Total Marks	Average Marks	Grade
Ravi	78	85	80	243	121.5	B
Priya	92	88	90	270	135	A
Arun	60	65	70	195	97.5	D
Sneha	88	95	92	275	137.5	A
Kiran	55	60	58	173	86.5	D
	74.6	78.6	78	MAX=275		
				MIN=173		
	Top Performer		Sneha			
	Weak Student		Kiran			

Names	Total Marks
Ravi	243
Priya	270
Arun	195
Sneha	275
Kiran	173



6. Results and Analysis

- Sneha was identified as the top-performing student
- Kiran was identified as the lowest-performing student
- Science showed relatively higher scores among students



7. Insights

- Data analysis helps identify student strengths and weaknesses
- Visualization makes comparison easier
- Performance trends can support decision-making

8. Conclusion

This project helped me understand the basics of data analysis using Excel and Python. It demonstrated how raw data can be transformed into meaningful insights using simple tools. It also improved my analytical and visualization skills.